

Lecture 12: Hypothesis testing in multiple regression

Economics 326 — Introduction to Econometrics II

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The model

- We consider the classical normal linear regression model:
 1. $Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_k X_{k,i} + U_i$.
 2. $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
 4. $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
 5. Conditional on $\mathbf{X}$, U_i 's are jointly normally distributed.
- We also assume **no perfect multicollinearity**: there exist no constants $c_1, \dots, c_k, c_{k+1}$ (not all zero) such that for **all** i 's:

$$c_1 X_{1,i} + \dots + c_k X_{k,i} + c_{k+1} = 0.$$

Testing a single coefficient

- Take the j -th coefficient β_j , $j \in \{0, 1, \dots, k\}$.
- Under our assumptions, conditional on $\mathbf{X}$, the OLS estimator $\hat{\beta}_j$ satisfies

$$\hat{\beta}_j \sim N(\beta_j, \text{Var}(\hat{\beta}_j | \mathbf{X})),$$

where $\text{Var}(\hat{\beta}_j | \mathbf{X}) = \sigma^2 / \sum_{i=1}^n \tilde{X}_{j,i}^2$ (see Lecture 11).

- Therefore, $(\hat{\beta}_j - \beta_j) / \sqrt{\text{Var}(\hat{\beta}_j | \mathbf{X})} \sim N(0, 1)$.
- Since σ^2 is unknown, so is $\text{Var}(\hat{\beta}_j | \mathbf{X})$. We estimate it by

$$\widehat{\text{Var}}(\hat{\beta}_j) = \frac{s^2}{\sum_{i=1}^n \tilde{X}_{j,i}^2},$$

where $s^2 = \sum_{i=1}^n \hat{U}_i^2 / (n - k - 1)$ (see Lecture 10).

- Conditional on $\mathbf{X}$,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_j)}} \sim t_{n-k-1}.$$

- Standard error: $\text{se}(\hat{\beta}_j) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_j)} = \sqrt{s^2 / \sum_{i=1}^n \tilde{X}_{j,i}^2}$.
- Consider testing $H_0 : \beta_j = \beta_{j,0}$ against $H_1 : \beta_j \neq \beta_{j,0}$.
- Under H_0 , conditional on $\mathbf{X}$,

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_j)}} \sim t_{n-k-1}.$$

- Let $t_{df,\tau}$ be the τ -th quantile of the t_{df} distribution.
- **Test:** Reject H_0 when $|T| > t_{n-k-1, 1-\alpha/2}$.
- **P-value:** $\text{p-value} = 2 \left(1 - F_{t_{n-k-1}}(|T|)\right)$, where $F_{t_{n-k-1}}$ is the CDF of the t_{n-k-1} distribution.

Testing a linear combination of coefficients

- Let $c_0, c_1, \dots, c_k, r$ be given constants. Consider testing

$$\begin{aligned} H_0 : c_0\beta_0 + c_1\beta_1 + \dots + c_k\beta_k &= r \text{ against} \\ H_1 : c_0\beta_0 + c_1\beta_1 + \dots + c_k\beta_k &\neq r. \end{aligned}$$

- **Example 1:** Consider $\ln Y_i = \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + U_i$. To test for constant returns to scale, $H_0 : \beta_1 + \beta_2 = 1$, set $c_0 = 0, c_1 = 1, c_2 = 1, r = 1$.
- **Example 2:** Consider $\ln(\text{Wage}_i) = \beta_0 + \beta_1 \text{Experience}_i + \beta_2 \text{PrevExperience}_i + \dots + U_i$. To test whether the two experience variables have the same effect on wage, $H_0 : \beta_1 - \beta_2 = 0$, set $c_0 = 0, c_1 = 1, c_2 = -1, c_3 = \dots = c_k = 0, r = 0$.
- **Example 3:** Consider $\ln(\text{Wage}_i) = \beta_0 + \beta_1 \text{Exper}_i + \beta_2 \text{Exper}_i^2 + \dots + U_i$. The marginal effect of experience is $\beta_1 + 2\beta_2 \text{Exper}_i$. For a concave profile ($\beta_2 < 0$), the marginal effect is smallest at the highest experience level. To test whether the marginal effect equals zero at $\text{Exper} = 20$: $H_0 : \beta_1 + 40\beta_2 = 0$, with $c_1 = 1, c_2 = 40, r = 0$.

Testing a linear combination of coefficients

- Conditional on $\mathbf{X}$, under $H_0 : c_0\beta_0 + c_1\beta_1 + \dots + c_k\beta_k = r$,

$$\begin{aligned} & \frac{c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k - r}{\sqrt{\widehat{\text{Var}}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k \mid \mathbf{X})}} \\ &= \frac{c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k - (c_0\beta_0 + c_1\beta_1 + \dots + c_k\beta_k)}{\sqrt{\widehat{\text{Var}}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k \mid \mathbf{X})}} \\ &\sim N(0, 1). \end{aligned}$$

- The variance of the linear combination is

$$\begin{aligned} & \text{Var}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k \mid \mathbf{X}) \\ &= \sum_{j=0}^k c_j^2 \text{Var}(\hat{\beta}_j \mid \mathbf{X}) + \sum_{j=0}^k \sum_{l \neq j}^k c_j c_l \text{Cov}(\hat{\beta}_j, \hat{\beta}_l \mid \mathbf{X}). \end{aligned}$$

Testing a linear combination of coefficients

- Consider

$$T = \frac{c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k - r}{\sqrt{\widehat{\text{Var}}(c_0\hat{\beta}_0 + c_1\hat{\beta}_1 + \dots + c_k\hat{\beta}_k)}}.$$

- Under $H_0 : c_0\beta_0 + c_1\beta_1 + \dots + c_k\beta_k = r$, conditional on $\mathbf{X}$,

$$T \sim t_{n-k-1}.$$

- **Two-sided test:** Reject H_0 when $|T| > t_{n-k-1, 1-\alpha/2}$.

CRS test: details

- Consider the model $\ln Y_i = \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + U_i$.
- We test for constant returns to scale: $H_0 : \beta_1 + \beta_2 = 1$.
- The test statistic: $T = \frac{\hat{\beta}_1 + \hat{\beta}_2 - 1}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_1 + \hat{\beta}_2)}}$.
- The estimated variance:

$$\begin{aligned} \widehat{\text{Var}}(\hat{\beta}_1 + \hat{\beta}_2) &= \widehat{\text{Var}}(\hat{\beta}_1) + \widehat{\text{Var}}(\hat{\beta}_2) \\ &\quad + 2\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2). \end{aligned}$$

- $\widehat{\text{Var}}(\hat{\beta}_1)$ and $\widehat{\text{Var}}(\hat{\beta}_2)$ are obtained from the standard errors reported by R.
- In R, `vcov(fit)` returns the full variance-covariance matrix of the coefficient estimates, including $\widehat{\text{Cov}}(\hat{\beta}_1, \hat{\beta}_2)$.
- Reject $H_0 : \beta_1 + \beta_2 = 1$ if $|T| > t_{n-3, 1-\alpha/2}$.

Example

- We generate 1000 observations from the model:

$$\left. \begin{aligned} L_i &= e^{l_i} \\ K_i &= e^{k_i} \end{aligned} \right\} \text{ where } l_i, k_i \text{ are iid } N(0, 1), \text{Cov}(l_i, k_i) = 0.5,$$

$$U_i \sim \text{iid } N(0, 1) \text{ is independent of } l_i, k_i,$$

$$Y_i = L_i^{0.35} K_i^{0.52} e^{U_i}.$$

- We estimate the equation:

$$\ln Y_i = \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + U_i.$$

- We test $H_0 : \beta_1 + \beta_2 = 1$ against $H_1 : \beta_1 + \beta_2 \neq 1$ at the 5% significance level.

```
set.seed(123)
n <- 1000
lnL <- rnorm(n)
lnK <- 0.5 * lnL + sqrt(1 - 0.5^2) * rnorm(n)
U <- rnorm(n)
lnY <- 0.35 * lnL + 0.52 * lnK + U
```

Example: regression output

- Regression output:

```
fit <- lm(lnY ~ lnL + lnK)
summary(fit)
```

Call:

```
lm(formula = lnY ~ lnL + lnK)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.8360	-0.6277	-0.0370	0.6538	3.3787

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.02093	0.03098	-0.676	0.499
lnL	0.31263	0.03735	8.371	<2e-16 ***
lnK	0.55176	0.03555	15.522	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9788 on 997 degrees of freedom

Multiple R-squared: 0.3942, Adjusted R-squared: 0.393

F-statistic: 324.4 on 2 and 997 DF, p-value: < 2.2e-16

- The variance-covariance matrix of the coefficient estimates:

```
vcov(fit)
```

	(Intercept)	lnL	lnK
(Intercept)	9.598283e-04	1.015829e-05	-4.491794e-05
lnL	1.015829e-05	1.394680e-03	-7.281792e-04
lnK	-4.491794e-05	-7.281792e-04	1.263649e-03

- The critical value $t_{n-3,0.975}$:

```
qt(1 - 0.025, df = fit$df.residual)
```

```
[1] 1.962346
```

Example: manual calculation

- From the regression output:

```
b1 <- coef(fit)["lnL"]
b2 <- coef(fit)["lnK"]
V <- vcov(fit)

cat("b1 =", b1, "\n")
```

```
b1 = 0.3126275
```

```
cat("b2 =", b2, "\n")
```

```
b2 = 0.5517621
```

```
cat("Var(b1) =", V["lnL", "lnL"], "\n")
```

```
Var(b1) = 0.00139468
```

```
cat("Var(b2) =", V["lnK", "lnK"], "\n")
```

```
Var(b2) = 0.001263649
```

```
cat("Cov(b1, b2) =", V["lnL", "lnK"], "\n")
```

```
Cov(b1, b2) = -0.0007281792
```

- The standard error of $\hat{\beta}_1 + \hat{\beta}_2$:

```
se_sum <- sqrt(V["lnL", "lnL"] + V["lnK", "lnK"] + 2 * V["lnL", "lnK"])
```

```
cat("se(b1 + b2) =", se_sum, "\n")
```

```
se(b1 + b2) = 0.03466944
```

- The test statistic:

```
T_stat <- (b1 + b2 - 1) / se_sum
```

```
cat("T =", T_stat, "\n")
```

```
T = -3.911526
```

- The critical value:

```
cv <- qt(1 - 0.025, df = fit$df.residual)
```

```
cat("|T| =", abs(T_stat), ", critical value =", cv, "\n")
```

```
|T| = 3.911526 , critical value = 1.962346
```

- Since $|T| > t_{997, 0.975}$, we reject H_0 .
- Ignoring the covariance leads to an incorrect result:

```
se_wrong <- sqrt(V["lnL", "lnL"] + V["lnK", "lnK"])
```

```
T_wrong <- (b1 + b2 - 1) / se_wrong
```

```
cat("T (ignoring covariance) =", T_wrong, "\n")
```

```
T (ignoring covariance) = -2.6302
```

Re-parametrization approach

- We test $\beta_1 + \beta_2 = 1$ in $\ln Y_i = \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + U_i$.
- Define $\delta = \beta_1 + \beta_2$, or $\beta_2 = \delta - \beta_1$, so that

$$\begin{aligned}\ln Y_i &= \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + U_i \\ &= \beta_0 + \beta_1 \ln L_i + (\delta - \beta_1) \ln K_i + U_i \\ &= \beta_0 + \beta_1 (\ln L_i - \ln K_i) + \delta \ln K_i + U_i.\end{aligned}$$

- Generate a new variable $D_i = \ln L_i - \ln K_i$.
- Estimate $\ln Y_i = \beta_0 + \beta_1 D_i + \delta \ln K_i + U_i$.
- Test $H_0 : \delta = 1$ against $H_1 : \delta \neq 1$.

Example: reparameterization

- Reparameterized regression output:

```
D <- lnL - lnK
fit2 <- lm(lnY ~ D + lnK)
summary(fit2)
```

Call:

```
lm(formula = lnY ~ D + lnK)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.8360	-0.6277	-0.0370	0.6538	3.3787

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.02093	0.03098	-0.676	0.499
D	0.31263	0.03735	8.371	<2e-16 ***
lnK	0.86439	0.03467	24.932	<2e-16 ***

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Residual standard error: 0.9788 on 997 degrees of freedom

Multiple R-squared: 0.3942, Adjusted R-squared: 0.393

F-statistic: 324.4 on 2 and 997 DF, p-value: < 2.2e-16

- The 95% confidence interval for the coefficient on $\ln K$:

```
confint(fit2, "lnK")
```

	2.5 %	97.5 %
lnK	0.7963561	0.932423

- The interval does not include 1, so we reject H_0 .
- In the original equation, $\hat{\beta}_1 + \hat{\beta}_2$ and $\text{se}(\hat{\beta}_1 + \hat{\beta}_2)$ equal the coefficient and standard error of $\ln K$ in the reparameterized regression.

Testing with `linearHypothesis()` in R

- The `car` package provides `linearHypothesis()` for testing linear restrictions on regression coefficients.
- Testing for constant returns to scale ($\beta_1 + \beta_2 = 1$):

```
library(car)
linearHypothesis(fit, "lnL + lnK = 1")
```

Linear hypothesis test:

```
lnL + lnK = 1
```

Model 1: restricted model

Model 2: $\ln Y \sim \ln L + \ln K$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	998	969.76				
2	997	955.10	1	14.657	15.3	9.793e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- `linearHypothesis()` reports an F-statistic. If $T \sim t_{n-k-1}$, then $F = T^2 \sim F_{1,n-k-1}$.
- For a single restriction, the F-test and the two-sided t-test are equivalent: $F = T^2$, with identical p-values.
- Testing for equal effects ($\beta_1 = \beta_2$):

```
linearHypothesis(fit, "lnL = lnK")
```

Linear hypothesis test:

lnL - lnK = 0

Model 1: restricted model

Model 2: lnY ~ lnL + lnK

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	998	968.42				
2	997	955.10	1	13.314	13.898	0.0002039 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1